MGT 404 Review Session Problems

Tax

A group of local activists in Cambridge, Massachusetts is trying to encourage lawmakers to expand the city's use of taxes to reduce consumption of products with negative externalities. Specifically, the group believes that the city should tax beer at restaurants, reasoning that higher beer prices would mean fewer drunk college students, who (they argue) make a lot of noise and upset downtown residents. The group favors imposing a tax of $2 per beer served at restaurants. For this question assume there is no baseline alcohol tax.

Cambridge restaurants provide beer according to the following weekly supply curve:

$$P_S(Q) = 2 + \frac{1}{5}Q,$$

and face the following weekly demand curve:

$$P_D(Q) = 12 - \frac{1}{5}Q,$$

where Q is in 1,000s of bottles per week.

a) What is the market equilibrium price of beer sold in a restaurant in Cambridge and the total quantity sold per week with no tax?

b) What would the new market equilibrium be if the beer tax were in place?

Price (inclusive of tax): _________

Quantity: ___________

c) What is the total tax revenue? What percent of the tax is borne by consumers?

Total tax revenue) ($): _______________

% paid by consumers ($): _____________

Production

Baking up some profits. Crumbs Bakery, which has been in the news lately, sells cupcakes through retail stores. The production of cupcakes takes both raw ingredients, and labor. The daily production function for cupcakes is given by

$$q = \frac{1}{2} \cdot r \cdot l + l^2 - 20$$

Where output is in dozens of cupcakes, r is pounds of raw ingredients, and l is hours of labor. The cost of a pound of raw ingredients is $p_r = \$1$ while the cost of an hour of labor is $p_l = \$12$.

(a) (8 points). What is the optimal number of pounds of raw ingredients to use per hour of labor?

Pounds of raw ingredients per hour of labor: ____________________

(Challenge Question) (b) (7 points). A standard day requires about 25 dozen cupcakes to be made. What is the cost-minimizing production plan for 25 dozen cupcakes? How much will this cost?

Pounds of raw ingredients: ____________________

Hours of labor: ____________________

Total cost: ____________________

Perfect Competition

Medical Devices. Jaysun Glove is a small manufacturer and distributor of medical examination gloves. The production process of such latex gloves is highly standardized and the industry is very competitive. Operating an assembly line to produce the gloves has a fixed cost of $20,000 per month, and the marginal cost of a case of 1,000 gloves increases with the scale of production. The total cost (in dollars) of producing q cases of gloves in a month and per assembly line is:

$$TC(q) \quad = \quad 20000 + 0.005 \cdot q^2$$

a. Find the marginal cost function, $MC(q)$.
$MC(q)$: ____

b. Find the average cost function, $AC(q)$.
$AC(q)$: ____

c. Assume that the market has come to a long-run (allowing for entry and exit) equilibrium. What is the price of a case of gloves?
Price: ____

Price Discrimination (1)

Pricing at the Taj Mahal. The Taj Mahal in Agra is the most popular monument in India, attracting thousands of visitors daily. The total daily demand for visits from foreign tourists is:

$$Q^F(p) = 4{,}000 - 200p,$$

where p is measured in US dollars. The Taj Mahal's operator, the Agra Development Authority (ADA), incurs a cost of $0.50 for every visitor who tours the Taj Mahal.

(a) (4 points). Find the profit-maximizing price that the ADA should charge to foreign tourists. How much profit does the ADA make every day?

Optimal price: ______________________________

Daily profit: ______________________________

(b) (2 points). There is also a large group of local residents who would like to see the monument. Total daily demand from locals is:

$$Q^L(p) = 20{,}000 - 20{,}000p.$$

At the price you found in part (a), how many locals would visit the Taj Mahal?

Number of locals: ______________________________

(c) (4 points). To promote access by the local community, the ADA moved towards charging local tourists – who have to show ID to qualify – a different admission price from the price it charges to foreigners.

Find the profit-maximizing price that the ADA should charge to local tourists.

Price: ______________________________

(d) How much additional profit does the ADA make from offering two different prices relative to the single-price strategy from part (a)?

Gain in Profit: ______________________________

(e) Out of the total visitors to the Taj Mahal, what share is made up of local residents?

Share of locals: ______________________________

Price Discrimination (2)

The Pennsylvania Ballet is preparing for its performance of "The Nutcracker", which attracts two different types of customers: one-time ballet visitors and ballet lovers. There are twice as many one-time visitors as there are ballet lovers. The marginal cost to the Ballet per ticket sold is \$20; assume that they have no capacity constraints and no preference for selling out any particular performance.

The Ballet conducted some market research and concluded that reservation prices for a single performance for the two groups are as follows:

	One-time Visitor (200 People)	Ballet Lover (100 People)
Afternoon performance (MC = \$20)	\$40	\$60
Evening performance (MC = \$20)	\$70	\$120

The profit-maximizing ticket pricing strategy, if the Ballet chooses a single price, is to:

The Ballet now considers charging different prices for afternoon and evening performances. The profit-maximizing choice has them:

Risk Aversion

Target. Target offers insurance for their digital camera sales, so if the camera breaks within two years of purchase, the insurance holder can receive a new camera for free. A new Canon EOS Rebel T5i runs $750 and has a 5% chance of breaking within the first two years of ownership.

a) What is the actuarially fair price of this insurance plan?

Price = ________________

Target charges $100 for the two-year insurance plan.

b) How much expected profit does Target make on each insurance plan sold (assuming their replacement cost is actually $750)? (Those of you who have shopped for electronics recently know that retailers must make a profit given how hard salespeople typically push these service plans.)

π = ________________

Target would say this high price is due to risk aversion. Assume customers have a utility function given by $\mathbf{ln}(\boldsymbol{w})$ where w is wealth. NOTE: When dealing with logarithms, rounding too soon can lead to numerical errors. As such, don't round until the end.

c) Would the consumer want the $100 insurance plan if their wealth (after buying the camera the first time) were $5,000 (assume they repurchase the camera if it breaks and they do not have insurance)?

YES NO

d) Assuming the **ln** utility function, what is this consumer's willingness to pay for the insurance plan? NOTE: To avoid rounding errors, when you compute any logarithms, please keep at least 6 digits after the decimal point.

WTP = __________

(Challenge Question) Critics say that the way Target justifies selling this insurance is by scaring customers into thinking that the probability of failure is much higher.

e) What probability of failure in the first two years would justify a price of $100 to someone who has wealth of $5,000 (after buying the camera the first time)?

Probability of Failure =______________